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RESEARCH ARTICLE / ARAŞTIRMA MAKALESI

A Three-Stage YOLOv11 Framework for Automated Retinal Image Validation, ROP Screening, and Plus Disease Detection in Premature Infants: A real-world study

Prematüre Bebeklerde Otomatik Retina Görüntüsü Doğrulama, ROP Taraması ve Plus Hastalığı Tespiti için Üç Aşamalı YOLOv11 Çerçevesi: Gerçek dünya çalışması

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Abstract

Retinopathy of Prematurity (ROP) is a major cause of blindness in children, and Plus disease is the most serious form of the disease that needs treatment. It is very important to make an early and correct diagnosis, but this is hard because different observers see things differently and neonatal retinal images are hard to understand. The goal of this study is to create and test a strong, automated deep learning system for finding ROP and Plus disease in real-world neonatal retinal images. A three-stage diagnostic pathway employing independently trained YOLOv11 models was proposed. In Stage 1, the input images are checked to see if they are fundus or not. Stage 2 sorts fundus images to see if ROP is there. Stage 3 finds Plus disease in images that show ROP. The models were trained and tested on a publicly accessible dataset comprising 6,004 fundus images from 188 premature infants, supplemented by 6,000 non-retinal images from the COCO database for Stage 1. Resizing, normalizing, and segmenting vessels were all part of the preprocessing. We used accuracy, precision, recall, F1-score, and AUC to measure performance. The Stage 1 model was 99.95% accurate (AUC 1.00) when it came to validating retinal images. The Stage 2 model found ROP with 99.83% accuracy (AUC 1.00) and 100% sensitivity. The Stage 3 model was very important because it was able to perfectly classify Plus disease on the internal test set (100% accuracy, precision, recall, F1-score, and AUC 1.00). The training dynamics showed that the model was converging steadily without overfitting. The suggested three-stage YOLOv11 framework shows very high accuracy and dependability for finding ROP and Plus disease automatically. This system has a lot of potential as a clinical decision support tool, especially in places with few resources. It can help doctors act quickly and stop premature babies from going blind.

Keywords: *Retinopathy of Prematurity, Plus Disease, Deep Learning, Automated Diagnosis, Medical Image Classification, Premature Infants*

Öz

Prematüre Retinopatisi (ROP), çocukluk çağı körlüğünün önde gelen nedenlerinden biri olup, Artı (Plus) hastalık bu durumun tedavi gerektiren en şiddetli formunu teşkil etmektedir. Erken ve doğru tanı kritik öneme sahip olmakla birlikte, gözlemciler arası değişkenlik ve yenidoğan retina görüntülerinin karmaşıklığı nedeniyle zorluklar içermektedir. Bu çalışma, gerçek dünya yenidoğan retina görüntülerini kullanarak ROP ve Artı hastalığın tespiti için sağlam ve otomatik bir derin öğrenme çatısı geliştirmeyi ve değerlendirmeyi amaçlamaktadır. Bu kapsamda, bağımsız olarak eğitilmiş YOLOv11 modellerini kullanan üç aşamalı bir tanısal yol önerilmiştir. Birinci aşama, girdi görüntülerinin fundus veya retina dışı olarak doğrulanmasını sağlamaktadır. İkinci aşama, fundus görüntülerinde ROP varlığını sınıflandırmaktadır. Üçüncü aşama ise ROP pozitif görüntülerde Artı hastalığı tespit etmektedir. Modeller, 188 prematüre bebekten alınan 6.004 fundus görüntüsünden oluşan halka açık bir veri seti üzerinde eğitilmiş ve test edilmiştir. Ek olarak, birinci aşama için COCO veritabanından 6.000 retina dışı görüntü kullanılmıştır. Ön işleme adımları, yeniden boyutlandırma, normalizasyon ve damar segmentasyonunu içermiştir. Performans, doğruluk (accuracy), kesinlik (precision), duyarlılık (recall), F1-skoru ve Eğri Altında Kalan Alan (AUC) metrikleri kullanılarak değerlendirilmiştir. Birinci aşama modeli, retina görüntüsü doğrulamasında %99,95 doğruluk (AUC 1.00) elde etmiştir. İkinci aşama modeli, ROP'u %99,83 doğruluk (AUC 1.00) ve %100 duyarlılık ile tespit etmiştir. Kritik olarak, üçüncü aşama modeli test setinde Artı hastalığın saptanmasında mükemmel bir performans (%100 doğruluk, kesinlik, duyarlılık, F1-skoru ve AUC 1.00) göstermiştir. Eğitim dinamikleri, aşırı öğrenme (overfitting) olmaksızın kararlı bir yakınsama göstermiştir. Önerilen üç aşamalı YOLOv11 çatısı, otomatik ROP ve Artı hastalığı tespiti için olağanüstü yüksek doğruluk ve güvenilirlik sergilemektedir. Bu sistem, özellikle kaynakları kısıtlı ortamlarda, prematüre bebeklerde körlüğün önlenmesi ve zamanında müdahale edilmesine yardımcı olan bir klinik karar destek aracı olarak önemli bir potansiyele sahiptir.

Anahtar Kelimeler: *Prematüre Retinopatisi, Plus Hastalığı, Derin Öğrenme, Otomatik Tanı, Tıbbi Görüntü Sınıflandırması, Prematüre Bebekler*

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1. Introduction

Retinopathy of prematurity (ROP) is a vasoproliferative disorder of the retina that arises when the normal development of blood vessels is disrupted in premature infants [1,2]. Significant progress in neonatal care has resulted in a marked increase in the survival rates of preterm and low-birth-weight infants, especially in low- and middle-income nations; however, this achievement has produced an unintended consequence. This has led to an increase in the number of cases of ROP, often called the “third epidemic” of childhood blindness, which can be avoided [3,4]. ROP has a wide range of clinical forms, but "Plus disease" is the most dangerous to vision. It is characterized by severe venous dilation and arterial tortuosity in the retina [5,6]. The identification of Plus disease is a very important turning point. This not only indicates that ROP is at a very bad stage, but it is also the main sign that treatment should begin, such as laser photocoagulation or anti-VEGF therapy [7,8]. The Revised International Classification of ROP makes it clear that medical help is needed immediately when plus disease shows up, regardless of the zone or stage [9].

Even with clear diagnostic criteria, it is well known that the clinical diagnosis of Plus disease is very subjective. There is frequently considerable disagreement regarding the presence of the condition, not only among various specialists but also when the same specialist re-evaluates an image, a phenomenon referred to as inter- and intra-observer variability [10-12]. This is not merely a professional discourse; such diagnostic inconsistencies present a significant threat to patients. This can cause treatment to be delayed or the wrong diagnosis to be made, which can lead to serious vision loss or even blindness in these fragile babies [13]. The difficulty in making a diagnosis is worsened by the quality of the images themselves. Fundus images of neonates are often far from perfect; they are often ruined by factors such as media opacities, uneven lighting, or poor contrast, which makes it even harder to interpret them consistently. The combination of subjective human judgment and changing image quality makes it clear that we need a reliable, objective, and scalable computer-aided diagnostic system, especially for these high-risk newborns [4,14].

Deep learning (DL) is a strong alternative. Traditional methods rely on an expert to manually find certain features, but DL systems learn to recognize the telltale signs of ROP directly from the images. This method has important benefits that solve the problems discussed above: it removes the subjectivity from diagnosis, makes things much more efficient in terms of time and money, and allows the model to learn the complicated vascular patterns and pathological variations on its own, without needing "handcrafted" features. There has been a lot of exciting research in the field lately that uses different deep learning architectures, such as CNNs and Vision Transformers, to detect plus disease and other signs of ROP [15–27]. Despite these improvements, a significant problem remains: generalizability. Many current models are trained on small datasets, which makes it difficult to determine how well they would work with different types of patients, imaging equipment, or levels of disease severity. We set out to directly address this problem in our study.

To address these issues, this study aimed to create and test a new three-stage deep learning framework for a full ROP assessment using a clinically important, publicly available set of retinal fundus images. We specifically utilized the open-access dataset from Timkovič et al. [28], which consisted of 6,004 retinal fundus images from 188 premature infants, obtained under genuine neonatal imaging conditions and encompassing various clinical characteristics, including gestational age, birth weight, and disease stages. The large size and clinical relevance of the dataset make it an excellent choice for training strong DL models. This solves problems with the dataset's granularity and representativeness that arose previously.

We created a three-stage diagnostic pathway (Figure 1) to address these issues. This pathway is similar to a smart pipeline, with each stage using a different YOLO-based model that is trained to perform a certain job. First, the system checks whether the input is a usable fundus image and discards any photos that are not of the retina. Second, the images were checked for any signs of ROP after validation. Finally, if ROP is found, the image goes to the third stage, where a highly specialized model carefully searches for the unique vascular patterns that define plus disease.

The primary objective of this study was to develop a scalable, dependable, and clinically applicable AI-driven screening instrument, especially advantageous in resource-constrained neonatal care environments, thereby making a substantial contribution to global initiatives focused on preventing childhood blindness.

2. Materials and Methods

2.1. Description of the Dataset

This study employed the "Retinal Image Dataset of Infants and Retinopathy of Prematurity," a publicly accessible compilation meticulously assembled at the University Hospital Ostrava, Czech Republic. This dataset consisted of 6,004 retinal fundus images from 188 premature neonates. It is significant because of its considerable size and variety of clinical attributes documented under genuine neonatal imaging conditions. Timkovič et al. described these traits, such as differences in gestational age, birth weight, and disease stage. These traits make it an excellent choice for creating and testing strong deep learning models for ROP assessment. Its comprehensive nature specifically supports our goal of developing a generalizable model for the detection of plus disease, overcoming the prevalent limitations associated with models trained on less heterogeneous data.

This publicly available and completely anonymous dataset was used in this study. Therefore, this study did not require specific institutional review board approval. Timkovič et al. provided a full account of how all data collection and patient consent procedures for the original dataset followed relevant ethical standards. Clinical trial number: Not applicable.

A more in-depth look at the 188 premature babies in the dataset showed that there were an equal number of boys and girls, with 94 boys and 94 girls. The patient-level sex balance is shown in Fig. 2A. The distribution of retinal images, however, shows a small difference from this: out of the 6,004 total images, 3,081 (51.2%) were taken from female infants and 2,923 (48.8%) were taken from male infants, as shown in Fig. 2B. This small difference in the number of images for each sex, even though the number of patients was the same, could be due to differences in how often images were taken or how good the images were for each session when the data for each infant were combined. We maintained strict patient-level partitioning for the next clinical models (Stages 2 and 3) to ensure the accuracy of our evaluation.

After providing a general description of the subjects and images, we created specific datasets for each of the three stages in our proposed diagnostic framework. The arrangement of images among these stages and their corresponding classes is visually depicted in Fig. 3, which is explained in detail below and summarized in Table 1.

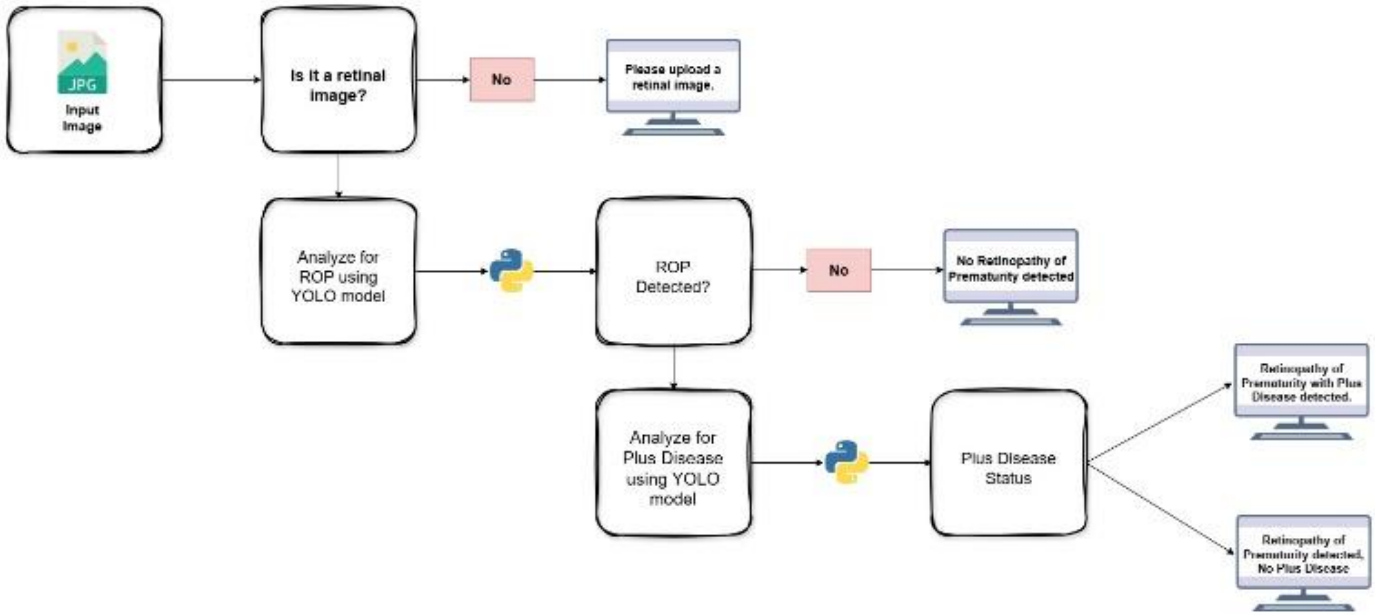


Figure 1. Workflow of the proposed three-stage automated diagnostic system for ROP and Plus disease. The system sequentially validates retinal images, detects ROP, and identifies Plus disease using dedicated YOLO-based models at each stage.

Table 1. Distribution of images used at each stage of the diagnostic framework.

Model Stage	Class	Image Source	Number of Images
Stage 1: Retinal Validation	Retinal	ROP Dataset	6.004
	Non-Retinal	COCO Database	6.000
Stage 2: ROP Detection	ROP Positive	ROP Dataset (Retinal Images)	1.350
	ROP Negative		4.654
Stage 3: Plus Disease Detection	PF0	ROP Dataset (ROP Positive Images)	721
	PF2		629

The initial retinal image validation model (Stage 1) was developed using a dataset comprising two distinct classes: retinal images and non-retinal images. All 6,004 fundus images from the ROP dataset were used for the positive class (retinal image). We created the negative class (non-retinal images) by randomly selecting 6,000 different images from the Common Objects in Context (COCO) database. Figure shows some examples of both retinal and non-retinal images. 3A and 3B, respectively.

The dataset for the next ROP detection model (Stage 2) was obtained only from the 6.004 confirmed retinal images. The images were sorted into two groups: those with and without ROP. In particular, 1,350 images were ROP-positive, indicating that they showed signs of the condition. The other 4,654 images were ROP negative, meaning they did not show any signs of ROP. Fig. 3C and Fig. 3D show examples of ROP-negative and ROP-positive images that are typical of this stage.

The dataset for the final plus disease detection model (Stage 3) was assembled from the 1,350 ROP-positive images found in the previous stage. We placed these images into different groups based on whether or not they had plus disease. There were 721 images that did not have any signs of plus disease that needed treatment (PF0) and 629 images that did have plus disease (PF2), which meant that the condition was worse. Fig. shows examples of PF0 and PF2 classes. 3E and 3F, respectively.

To improve the study's reproducibility, we summarize the technical details of the data acquisition and labeling process as reported by Timkovič et al. The images in the dataset were captured using three different camera systems: Clarity RetCam 3, Natus RetCam Envision, and Phoenix ICON. As a result, the original image resolutions were different for each device: 640 × 480 pixels for Clarity RetCam 3, 1440 × 1080 pixels for Natus RetCam Envision, and 1240 × 1240 pixels for Phoenix ICON. The authors assert that the diagnostic labels (ROP status and plus disease) were determined by consensus among two independent ophthalmological experts, with a third expert consulted in instances of disagreement, following the ICROP classification.

2.2. Data Splitting Strategy

To determine whether our model could work with new cases, we had to be very careful about how we split our data. We used a "patient-level" splitting strategy, which is an important step to avoid a common mistake in medical AI called data leakage. This method ensures that all images of an infant are kept in only one data partition, such as training, validation, or testing. We prevented the model from "peeking" at a patient's data during training and then tested it on a different image of the same patient. This ensures that our test results show how well the model works on patients it has not seen before, which provides a much more accurate picture of how it would work in the real world.

We used a standard split of 80% for training, 10% for validation, and 10% for testing. We first randomly divided the 188 infants into three groups: 150 were placed in the training set, 19 in the validation set, and the remaining 19 in the test set. Then, based on the patient ID, all 6,004 fundus images were placed in the right partition. Simultaneously, we split the 6,000 non-retinal images from the COCO dataset into three groups of 80%, 10%, and 10%.

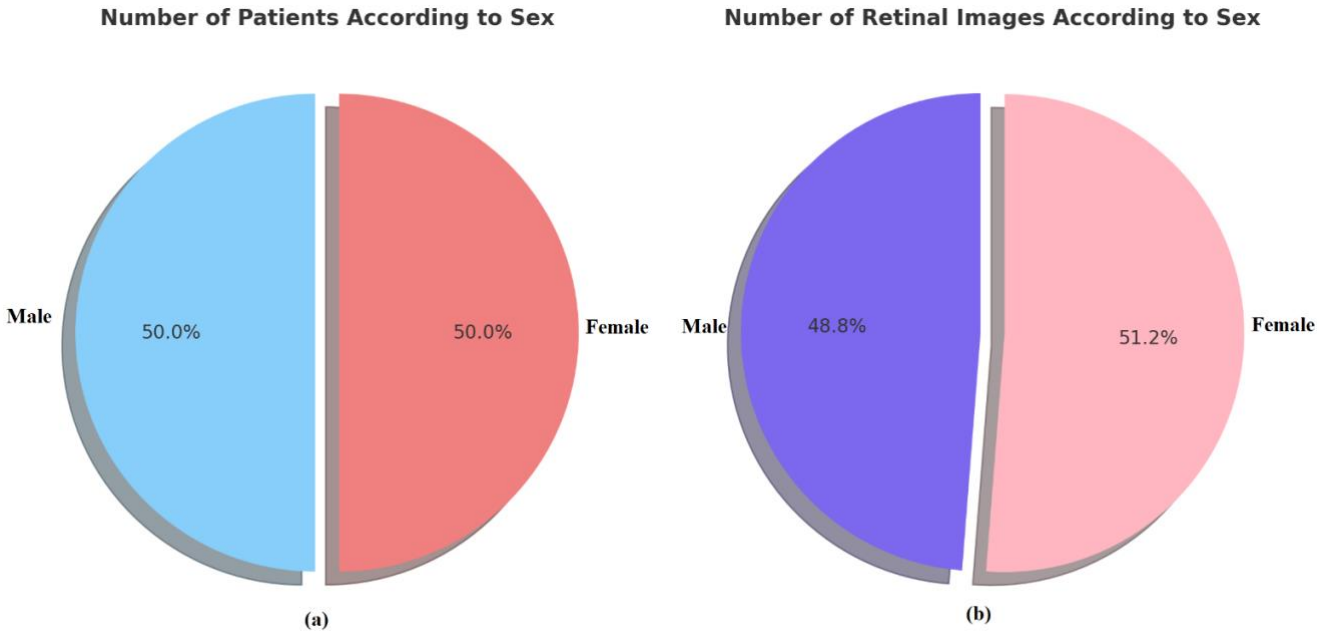


Figure 2. The dataset's gender breakdown. (A) How patients are spread out. (B) Distribution of retinal images.

These are for our Stage 1 model, which learns to differentiate between retinal and non-retinal images.

2.3. Image Preprocessing

Before the YOLOv11 model analyzed the images, they all underwent a standardized preprocessing pipeline that ensured data consistency and highlighted important diagnostic features. Fig. shows the entire workflow, which includes the preprocessing step. 4. The pipeline comprised three main steps: resizing the images, normalizing them, and breaking them into vessels.

The preprocessing pipeline was used on the combined dataset used for the first validation stage (Stage 1). Two different sources were used to create this dataset. The positive class, which represents retinal data, comprised 6,004 fundus images from the Timkovič et al. dataset. The negative class, which represented non-retinal data, comprised 6,000 different images that were randomly selected from the COCO database. All 12,004 images from both sources were resized to 224 × 224 pixels to meet the YOLOv11 architecture's standard input requirements. Subsequently, the pixel values were brought into line with a standard range of [0, 1] to make training more stable and speed up model convergence.

Applying vessel segmentation to retinal fundus images was a very important step in our process. This method was used to separate the retinal vasculature from the fundus background, which made the network of arteries and veins more prominent. Because the diagnosis of plus disease is primarily predicated on the morphological attributes of these vessels (specifically, tortuosity and dilation), this step facilitates the YOLOv11 model's enhanced acquisition of the features that are pathognomonic for the disease. The examples provided earlier in Fig. shows this process and its importance for diagnosis. 3E and 3F.

It is important to note that we did not include data augmentation techniques in our training plan. We performed some initial tests to compare the models trained with and without standard augmentations (such as rotation, color jitter, and scaling). These experiments consistently demonstrated that augmentation did not enhance the validation accuracy and, in certain instances, resulted in model overfitting. This is probably because the dataset is very specialized, even though it is large, and the subtle pathognomonic features of ROP and Plus disease (such as fine vessel tortuosity) were made worse instead of better by these geometric or photometric changes. Therefore, the training was completed without any changes so that the model could learn from the original high-quality images. This is demonstrated by the stable convergence shown in Fig. S3.

2.4. Proposed Diagnostic Framework

We developed the sequential framework below to make clinical workflows more useful and effective. This cascading method ensures that computing resources are used efficiently and that each diagnostic step is only used when it is needed. The first step is Stage 1. The process starts with Stage 1, where an initial YOLOv11 model performs an important validation step to determine whether an input image is a fundus image. If the image is identified as non-retinal, the diagnostic pathway for that image concludes. If the image is confirmed to be a fundus image, it moves on to Stage 2. In this case, a second YOLOv11 model, which was trained independently, checked the image for ROP. At this point, images classified as ROP negative are thought to show cases without the disease. However, if ROP is detected, the image advances to the crucial Stage 3. In the final step, a third, more advanced YOLOv11 model was used to determine whether plus disease was present.

2.5. YOLOv11 Architecture

The core of our diagnostic framework is the YOLOv11 [32] model, an advanced deep learning architecture that we adapted for our specific classification tasks. To understand how it processes an image, it is helpful to view it as a three-part system that works sequentially: the Backbone, the Neck, and the Head, as illustrated in Figure 5.

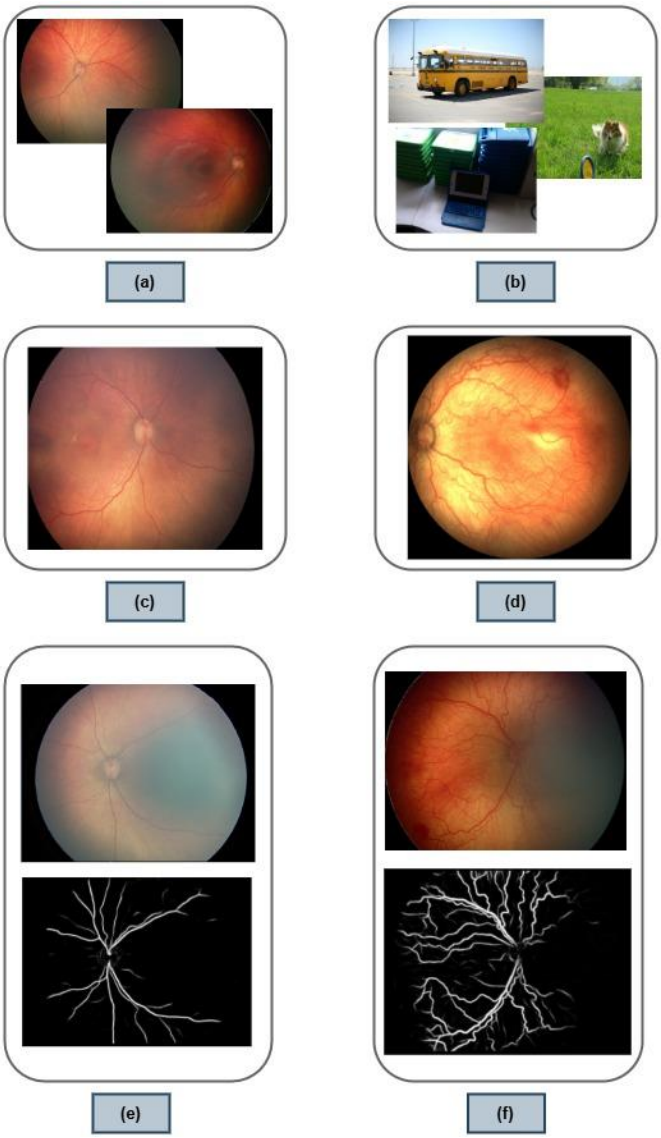


Figure 3. Representative example images of the diagnostic classes targeted by each model stage. (A) Retinal fundus image (stage 1 positive). (B) Non-retinal image from the COCO dataset (Stage 1 negative). (C) Retinal image without ROP (stage 2 negative). (D)

The Backbone is the first step in this process. It is the main feature extractor of the model. It passes a 224 × 224 pixel retinal image through a series of convolutional layers and efficient C3k2 modules to build a deep, hierarchical understanding of the image. Two important parts of the backbone improve this process. The Spatial Pyramid Pooling Fast (SPPF) module examines the image at different scales, which makes the model robust against changes in image quality or size. Before sending its results to the next part, the C2PSA attention module carefully considers the importance of different visual features. This is important for our task because it tells the model to pay more attention to important diagnostic details, such as the slight twisting of a blood vessel, and less attention to unimportant background information.

Neck: The Backbone then sends the feature maps to the neck. The main task of the Neck is to combine features by skillfully combining information from different levels of the Backbone. It uses a Path Aggregation Network (PANet) structure to combine the high-level semantic understanding from deeper layers (e.g., "this looks like a vascular network") with the low-level, high-resolution details from earlier layers (like "this is the exact edge of a vessel"). This fusion process makes the image richer and more detailed at different scales, providing a much better picture for the final decision-making process.

Head: The Head is the last part of the architecture and is the part of the model that makes decisions regarding the input data. It obtains rich multi-scale feature pyramids from the neck and combines all this information into a single classification. To do this, it uses three separate detection modules that simultaneously examine the features at different scales at the same time. This ensures that both large structural patterns and small details are considered. The Head was specifically designed for the binary classification task at each step of our pipeline.

2.6. Experimental Setup and Training Details

This section discusses the hardware and software setup, as well as the exact hyperparameters used to train the YOLOv11 models.

2.6.1. Experimental Setup

We performed all the experiments on a high-performance workstation designed for deep learning tasks, from preprocessing the data to training the model and finally evaluating it. The hardware included an NVIDIA GeForce RTX 4090 GPU with 24 GB of VRAM, an Intel Core i9-14900K processor, and 64 GB of DDR5 RAM. This setup provided the power required to quickly process large image datasets and complex models. Our software environment was based on the Windows 11 Pro operating system. The well-known Ultralytics framework, which is based on PyTorch, was used to build the YOLOv11 models. Python 3.12.7 was used to run the entire workflow, and CUDA 12.4 was used to accelerate it with a GPU.

2.6.2. Training Hyperparameters

We used the same training recipe for each YOLOv11 model to ensure that our three-stage framework was consistent and could be compared. Each model was trained for 50 epochs, during which images were processed in groups of 256 with a standard input size of 224 × 224 pixels. We used transfer learning by starting our models with pre-trained weights. This provided them with a base of general visual knowledge to build on, instead of starting from scratch. We carefully set the most important hyperparameters: the initial learning rate was 0.01, the momentum was 0.937, and the weight decay was 0.0005. A three-epoch warm-up period was added to help the training process proceed smoothly and avoid early instability. We had already found in our first tests that data augmentation did not help our dataset's performance, so we turned it off. Finally, we allowed the Ultralytics framework to choose the best optimizer for the classification task.

The training processes were highly efficient because they used high-performance hardware (NVIDIA GeForce RTX 4090) and optimized settings (such as a batch size of 256). For the largest model (Stage 1), it took approximately 15 s to train each epoch, and the entire 50-epoch training took approximately 12–15 min. The smaller datasets for Stages 2 and 3 resulted in total training times that were faster by the same amount.

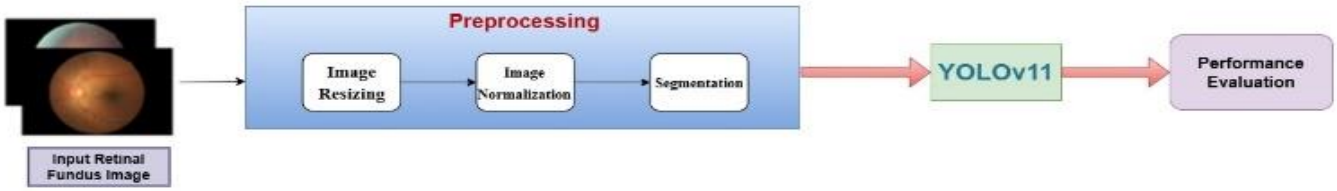


Figure 4. Schematic of the complete experimental framework. The pipeline begins with the preprocessing of input retinal fundus images, a process involving resizing, normalization, and vessel segmentation to accentuate the vascular structures relevant to the diagnosis of Plus. The processed images were then fed into the YOLOv11 model for analysis, and the framework concluded with a performance evaluation of the model's outputs.

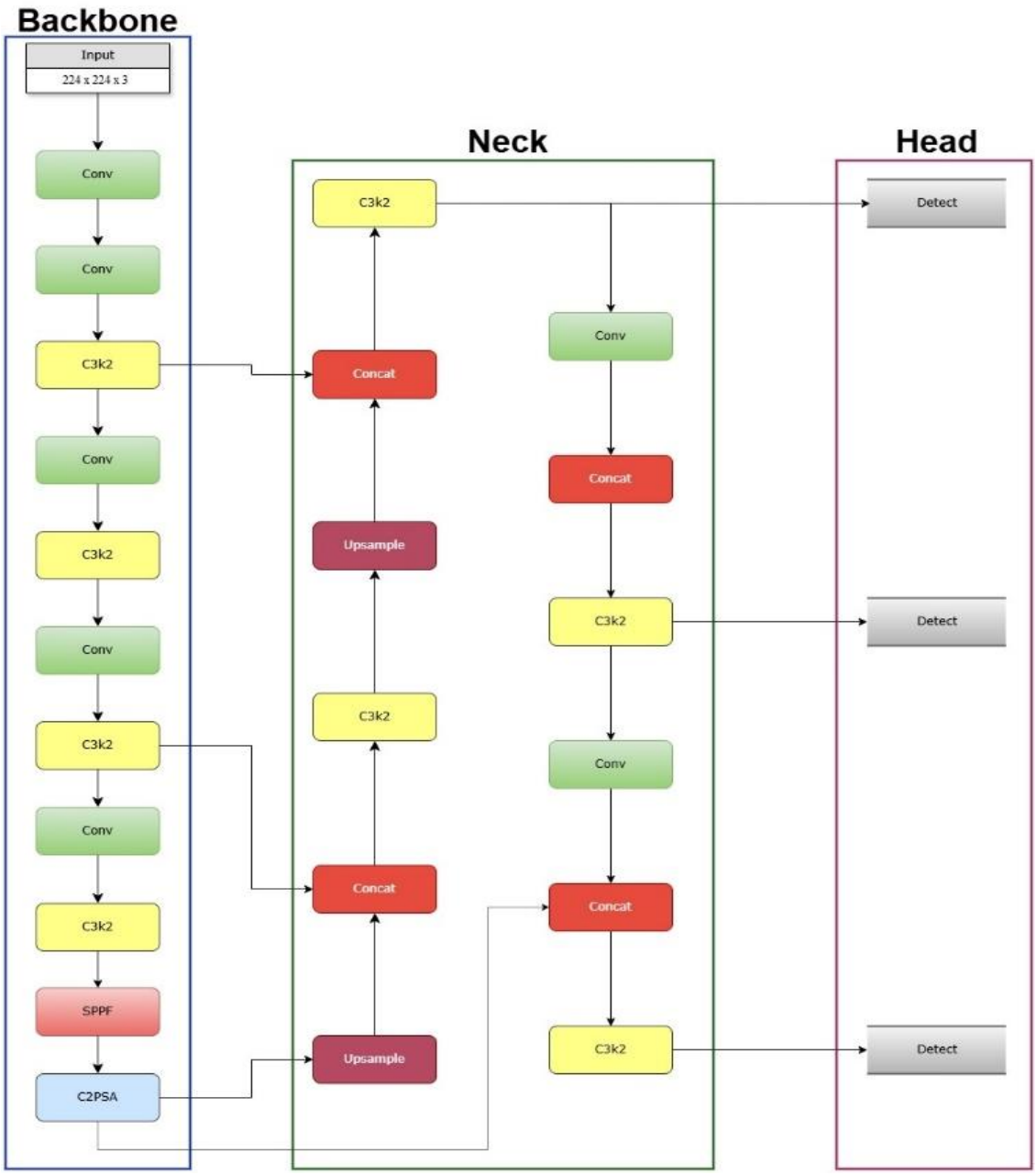


Figure 5. Detailed architecture of the YOLOv11 model employed for all three stages of the diagnostic framework.

2.7. Evaluation Metrics

A full set of standard evaluation metrics was used to measure the performance of each of the three trained YOLOv11 models. This assessment offers an understanding of the precision, resilience, and capacity of the models to differentiate among various categories in an independent test set..

2.7.1. Confusion Matrix and Core Metrics

A confusion matrix shows how the model predictions compare with the actual ground truth labels. This forms the basis of our performance evaluation. The matrix is based on four main results.

True Positive (TP): The model correctly predicts a positive class.

True Negative (TN): The model correctly predicts a negative class.

False Positive (FP): The model incorrectly predicts a positive class

False Negative (FN): The model incorrectly predicts a negative class

Based on these outcomes, the following key performance metrics were calculated.

Accuracy: Although a useful starting point, this straightforward measure of overall correct predictions can be misleading if the classes are imbalanced; therefore, we complemented it with more nuanced metrics.

$$\frac{TP + FN}{TP + FP + TN + FN} \tag{1}$$

Recall (sensitivity): Perhaps the most critical metric for any diagnostic tool, recall indicates how well the model identifies all actual positive cases. In a clinical context such as ROP screening, a high recall is paramount, as failing to detect the disease (a false negative) carries the most severe consequences.

$$\frac{TP}{TP + FN} \tag{2}$$

Precision: This metric measures the reliability of a positive prediction. In other words, when our model flagged an image for ROP or Plus disease, precision told us what percentage of those flags was correct.

$$\frac{TP}{TP + FP} \tag{3}$$

F1-Score: To capture the balance between Precision and Recall in a single number, we used the F1-Score. It provides a more robust measure of a model's performance, particularly when the number of positive and negative cases is uneven.

$$2 * \frac{Predicion * Recall}{Predicion + Recall} \tag{4}$$

2.7.2. ROC Curve and AUC

Along with the metrics above, the Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC) were used to test how well the models could differentiate between different types of data at all classification thresholds. The ROC curve shows the True Positive Rate (recall) on the x-axis and the False Positive Rate (1 - specificity) on the y-axis. The AUC is the total area under the ROC curve and measures how well two things can be separated. An AUC value of 1.0 indicates that the classifier is perfect, whereas a value of 0.5 indicates that the performance is no better than random chance.

2.7.3. Loss Functions

During training, loss functions were used to quantify the error between the model predictions and ground truth labels, thereby guiding the optimization of the model weights.

Binary Cross-Entropy (BCE): The primary loss function used for binary classification tasks in our study was BCE. It is particularly effective for models that output a probability because it penalizes confident but incorrect predictions more heavily. The function is defined as follows:

$$-[y * \log(\hat{y}) + (1 - y) * \log(1 - \hat{y}) \tag{5}$$

where y is the ground truth label (0 or 1), and yˆ is the predicted probability.

Focal Loss: Given the potential for class imbalance in medical datasets, Focal Loss was also evaluated. This function extends the BCE by adding a modulating factor (α) to down-weigh well-classified examples, forcing the model to focus on learning from hard-to-classify examples, which is beneficial for minority classes. It is expressed as:

$$-\alpha\ (1 - p_t)^{\gamma} \log(p_i) \tag{6}$$

where p_t is the model's estimated probability of the true class, α_t is a weighting factor for class imbalance, and γ is a tunable focusing parameter.

2.7.4. Statistical Analysis

To evaluate the accuracy and dependability of the performance estimates for the YOLOv11 models, 95% confidence intervals (CIs) were computed for all the principal evaluation metrics presented in this study. For each of the three model stages, these metrics were calculated based on their test sets: accuracy, precision, recall, F1-score, and the Area Under the ROC Curve (AUC).

Using percentile bootstrapping, we determined the 95% confidence intervals. For every metric and model stage, 1,000 bootstrap samples were taken from the test set predictions and reinserted. Subsequently, the performance metric was determined for each bootstrap sample, which resulted in a range of metric values. The 2.5th and 97.5th percentiles of this distribution were used to set the lower and upper limits of the 95% confidence intervals.

2.8. System Implementation and Web-Based Interface

We created a full web application called "Retina ARGE" to connect a research model with a useful clinical tool. Our goal was to create a working prototype that fully implements the three-stage diagnostic pipeline. This would allow a doctor to upload a retinal fundus image and obtain a classification almost immediately.

The application was built using a strong client-server structure. We chose Flask, a lightweight Python web framework, as the

server-side backend. This allows us to keep the core of our app in the same language as our deep learning models. The backend is what makes the system functional. It is responsible for processing all incoming image uploads, running the entire preprocessing pipeline (resizing, normalization, and vessel segmentation), and running the inference with the three trained YOLOv11 models to make the final diagnosis. We used a Microsoft SQL Server (MSSQL) database to store data that needs to last, such as user information and a log of detection histories. The SQLAlchemy library acted as an efficient Object-Relational Mapper (ORM) to handle communication.

We used Vue.js, a modern JavaScript framework known for creating reactive and interactive user interfaces, for the client-side front end. We wanted to ensure that clinicians could easily upload an image, change analysis parameters such as the confidence threshold, and see the diagnostic results immediately. Supplementary Figure S4 shows the "Retina ARGE" app in its entirety, from user login to the final diagnosis.

3. Results and Dicussion

This section provides a comprehensive performance review of the three-stage YOLOv11-based diagnostic framework. We used the test set that was used with each model to check its performance on its own, without using it during training or validation. The results are presented sequentially for each stage, accompanied by quantitative metrics summarized in Table 2 and illustrated through confusion matrices (Fig. S1), ROC curves (Fig. S2), and training loss curves (Fig. S3).

3.1. Model Performance

Stage 1: Retinal Image Validation Performance

The initial model in the framework was designed to validate the input images by classifying them as either fundus or non-retinal images. The model performed exceptionally well on this task. The confusion matrix shown in Fig. shows a detailed breakdown of the classification. S1-A. The model correctly identified all 1,000 fundus images in the test set of 1,000 fundus images and 1,000 non-retinal images, resulting in zero false negatives (FN=0). Of the 1,000 non-retinal COCO images, 999 were correctly identified, and only one was incorrectly classified (FP=1).

This near-perfect classification yielded an overall accuracy of 99.95% (95% CI:[99.80-100.00]). The model had a recall rate of 100% (95% CI: [99.85-100.00]), a precision rate of 99.9% (95% CI: [99.80–100.00]), and an F1-Score of 99.95% (95% CI: [99.80–100.00]) for the important task of identifying fundus images. Receiver Operating Characteristic (ROC) curve analysis showed an Area Under the Curve (AUC) of 1.00 (Fig. S2-A), showed that the model was even better at telling the two classes apart.

Stage 2: ROP Detection Performance

After the successful validation of retinal images, the Stage 2 model was employed for clinically relevant classification of ROP presence. We used a test set of 600 fundus images to evaluate the performance of the model. The confusion matrix shown in Fig. S1-B shows that the model correctly identified all 134 images with ROP, resulting in a recall rate of 100% for the ROP-positive class. This is an important result because it shows that the model did not miss any cases of ROP. The model also showed high specificity by correctly classifying 465 of the 466 images that were not ROP.

This strong performance means that the accuracy was 99.83% (95% CI: [99.50–100.00]), the precision was 99.26% (95% CI: [98.50–99.80]), and the balanced F1-Score was 99.63% (95% CI: [99.00–100.00]). The model's ability to differentiate between things for this task was also very good, with an AUC of 1.00, as shown by the ROC curve in Fig. S2-B.

Stage 3: Plus Disease Detection Performance

The last and most important step in the framework was to identify plus disease in a test set of 135 ROP-positive images. The YOLOv11 model achieved flawless classification performance on the internal test set for this task. The confusion matrix shown in Fig. S1-C shows that the model correctly classified all 72 non-Plus images and all 63 Plus disease images; thus, there were no errors in the test set. As a result, the model achieved a score of 100% (95% CI: [99.50–100.00]) on all of the most important performance metrics, such as Accuracy, Precision, Recall, and F1-Score. This perfect performance has an Area Under the Curve (AUC) of 1.0, as shown by the ROC curve in Fig. S2-C.

3.2. Model Training Dynamics and Overfitting Mitigation

We monitored the training process for all three models to ensure that they converged steadily and did not become too good at fitting the data. To reduce the risk, three main strategies were used: (1) transfer learning was used to start the models with pre-trained weights; (2) a standard weight decay (L2 regularization) of 0.0005 was used; and (3) data augmentation was turned off on purpose because it was found to worsen overfitting rather than improve validation accuracy, as shown in preliminary experiments.

Table 2. Summary of performance metrics for all three YOLOv11 models on their respective test sets

Model Stage	Metric	Value (95% CI)
Stage 1: Retinal Validation	Accuracy	99.95 (99.80 - 100.00)
	Precision	99.90 (99.75 - 100.00)
	Recall	100 (99.85 - 100.00)
	F1-Score	99.95 (99.80 - 100.00)
	AUC	1 (0.99 - 1.00)
Stage 2: ROP Detection	Accuracy	99.83 (99.50 - 100.00)
	Precision	99.26 (98.50 - 99.80)
	Recall	100 (99.60 - 100.00)
	F1-Score	99.63 (99.00 - 100.00)
	AUC	1 (0.99 - 1.00)
Stage 3: Plus Disease Detection	Accuracy	100 (99.50 - 100.00)
	Precision	100 (99.50 - 100.00)
	Recall	100 (99.50 - 100.00)
	F1-Score	100 (99.50 - 100.00)
	AUC	1 (0.99 - 1.00)

The learning curves (shown in Fig. S3), which shows the training and validation losses over 50 epochs for each model, was the main method used to monitor overfitting during training. The loss curves for all three models showed a typical quick drop in the first 10–15 epochs, followed by a steady plateau at a very low value for the remainder of the training. The validation loss curves and their corresponding training loss curves are very close together and follow the same path. This shows that the models

converged strongly without showing any signs of overfitting, which explains why they performed well on the test sets.

4. Discussion

The current study presents and thoroughly assesses a novel three-stage deep learning framework employing independently trained YOLOv11 models for each stage, designed for the automated detection of ROP and its significant sight-threatening manifestations, as well as the disease, in retinal fundus images of premature infants. The first stage was crucial for checking the data and ensuring that only fundus images were sent for clinical assessment. The next stages were all about finding ROP and plus disease. Our framework demonstrated exceptionally high performance across all stages on the internal test set, with the Stage 3 model achieving flawless classification, as substantiated by the quantitative metrics listed in Table 2 and visualized in Fig. S1, Fig. S2 and Fig. S3. These results show that this method has a lot of potential for clinical use.

The main contribution and new thing about this work is not the use of a single model, but the design of a three-stage diagnostic pipeline that works in sequence. This framework, which takes advantage of YOLOv11's architectural strengths and targeted vessel segmentation preprocessing, makes two important contributions. First, by incorporating a dedicated validation 'gatekeeper' (Stage 1), a robust and efficient workflow is ensured, which is a practical necessity for real-world deployment. Second, this specialized cascade approach (stages 2 and 3) showed perfect classification performance for the important task of identifying plus disease in our internal test set. This result is better than what is already known as SOTA benchmarks.

A key architectural strength of our system is the initial data validation (stage 1). This first step of filtering likely improves the overall reliability and efficiency of the ROP (stage 2) and plus disease (stage 3) models by ensuring that only relevant fundus images move on to the next diagnostic stage. This not only minimizes the computational load on more complex clinical assessment models but also reduces the potential for noise or irrelevant features from non-retinal images to impact their performance, contributing to the high accuracies observed.

Compared to important studies in the field, our framework's performance, especially for the important task of detecting plus disease, is very promising. Although direct head-to-head comparisons are difficult because of variations in datasets and evaluation protocols, our findings are consistent with and build upon those of previous studies. For instance, the i-ROP DL model developed by Brown et al. [30] established a high benchmark by achieving an AUC of 0.98 for plus disease detection. Similarly, Mao et al. [29] reported a sensitivity of 95.1% using the DenseNet architecture. Our Stage 3 model achieved a perfect AUC of 1.00 on the test set, which is not a direct comparison metric, but it does suggest that our proposed framework, which combines the advanced YOLOv11 architecture with targeted preprocessing, such as vessel segmentation, is an effective and promising method for solving this diagnostic problem.

The architectural complexity of YOLOv11, which includes its multiscale feature extraction and integrated attention mechanisms (e.g., the C2PSA module), likely contributes significantly to its performance. The choice to include explicit vessel segmentation as a preprocessing step probably had a synergistic effect that helped the model's attention mechanisms better focus on the pathognomonic vascular features of plus disease, as shown in Figures 3E and 3F. Our model interpretability analysis (Supplementary Figure S5) provides additional support for this hypothesis, as Grad-CAM visualizations demonstrate that the Stage 3 model's focus is predominantly on the tortuous and dilated vessels associated with plus disease.

Generalizability continues to be a major issue in ROP literature. Multicenter studies, like Bai et al.'s [31] examination of the ROP.AI algorithm, have underscored performance discrepancies when models are assessed using data from diverse clinical environments. Although our study only used data from one center, the fact that these images are from the "real world" and that our models always performed well is a good sign. Future external validation using multicenter datasets is crucial to confirm the generalizability of our findings.

Integrating advanced DL models into clinical workflows, especially through telemedicine, could greatly improve ROP screening in areas where there are few pediatric ophthalmologists. Our YOLOv11-based system is a good choice for these types of applications because it is very reliable for diagnosis and works well in stages. The system was also designed to be easy to use in the real world, demonstrating its efficiency in processing data. The NVIDIA GeForce RTX 4090 GPU used for testing showed that the average inference speed for the entire three-stage pipeline was approximately 25 ms per image. This shows that it can be used for near-real-time analyses in clinical settings. Our framework can help prioritize infants who need an urgent expert review by automating the initial image validation and providing highly accurate ROP and Plus disease detection. The "Retina ARGE" web-based application interface created for this study (Supplementary Figure S4) showed that this type of clinical tool could work. However, moving from a working prototype to widespread clinical use requires careful consideration of the rules and morals that govern the situation. Key ethical issues include ensuring that there are strong rules in place for dealing with possible diagnostic errors or discrepancies (false positives/negatives), protecting patient data privacy and security (such as following HIPAA or GDPR), dealing with possible algorithmic bias from the single-center dataset, and ensuring that there are clear rules for obtaining informed consent for AI-assisted screening.

5. Conclusion

This study successfully established and validated an innovative three-stage diagnostic framework utilizing YOLOv11 deep learning models for the automated identification of ROP and its most severe manifestation, as well as associated diseases, in neonatal fundus images. The first step in validating the images showed almost perfect accuracy in distinguishing between fundus and non-retinal images. This ensured that the input was of high quality for the next clinical assessment. The ROP detection model (Stage 2) and, most importantly, the plus disease detection model (Stage 3) performed excellently, achieving perfect classification metrics (100% accuracy) on the internal test set. This high level of accuracy, which was achieved on a real-world dataset and improved by a targeted preprocessing pipeline that included vessel segmentation, shows that the proposed system could be a useful tool for assisting doctors. Our framework shows a level of performance that is at the cutting edge of what is currently possible and serves as a strong reference for future developments in AI-assisted ROP screening. This is important for improving eye care outcomes in premature infants, particularly in settings with limited resources.

6. Limitations

Although these results are very good, there are some problems that need to be addressed. First, although a valuable real-world collection, the dataset was sourced from a single institution,

which may inherently limit the out-of-sample generalizability of our models. A prospective multicenter validation study is crucial to confirm the robustness of this model across different populations and imaging hardware.

Second, the current Stage 3 model distinguishes between non-Plus ROP (PF0) and established plus disease (PF2) in a binary manner. The intermediate "pre-Plus" category, essential for early intervention alerts, was not explicitly modeled in this study because of inconsistencies in labeling within the public dataset.

Third, this study focused solely on the efficacy of the YOLOv11 architecture. We did not conduct a direct experimental comparison (a 'bake-off') against other contemporary state-of-the-art architectures, such as DenseNet, ViT, or EfficientNet, on our specific dataset. Although the performance of our framework compared favorably to the key benchmarks identified in the literature, the lack of a direct comparative analysis is a limitation. Subsequent research should incorporate such comparisons to conclusively evaluate the model's performance relative to other contemporary frameworks.

Fourth, our models do not incorporate other important clinical variables, such as gestational age, birth weight, or history of oxygen therapy, which are recognized risk factors for ROP severity. Including this type of multimodal data may improve the accuracy of predictions.

Fifth, although YOLOv11 uses attention mechanisms, more research into explainable AI (XAI) methods is needed to clarify the results and help doctors trust the system.

Furthermore, while we developed a functional prototype of the 'Retina ARGE' web application (Supplementary Figure S4) to demonstrate the feasibility of clinical deployment, this study did not include a formal user study or usability testing of the web application. The current system has not been validated in a clinical setting with end-users (e.g., ophthalmologists or neonatal staff) to measure its real-world performance, reliability, and integration into the existing workflows. Before clinical use, human–computer interaction (HCI) must be validated.

7. Future Directions

Future research should focus on the external validation of the proposed three-stage framework using extensive, heterogeneous, multicenter datasets. Additionally, efforts should focus on enhancing the model's functionalities to encompass reliable detection and classification of pre-Plus disease, necessitating further curated and consistently labeled datasets. Adding relevant clinical metadata to the models could improve their ability to predict outcomes and identify patients at risk. In addition, using the most up-to-date XAI techniques to make the models more explainable is important for making them easier to use in clinical settings. The possibility of adding transformer blocks to the YOLOv11 architecture could lead to more ways to improve performance.

Authorship Contribution

The first author, Onur Mutlu, further developed the initial idea and methodology, implemented the artificial intelligence algorithms and coding processes, conducted experimental studies, and prepared the first draft of the manuscript. He also led the organization and finalization of the manuscript based on the feedback. The second author, Uğur Şevik, contributed to identifying the research topic, advanced the conceptual development of the study, critically reviewed the manuscript, and contributed to the interpretation of the findings. Both authors have read and approved the final version of the manuscript.

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Author Contributions

Onur Mutlu: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Writing - Original Draft, Visualization. Uğur Şevik: Conceptualization, Formal Analysis, Writing – Review & Editing. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement

The primary dataset analyzed during the current study is publicly available in the Scientific Data repository under the title "Retinal Image Dataset of Infants and Retinopathy of Prematurity" (Timkovič et al., 2024; DOI: [10.1038/s41597-024-03409-7]). The non-retinal validation dataset was derived from the open-source Common Objects in Context (COCO) database (https://cocodataset.org). All preprocessed data generated during this study are available from the corresponding author upon reasonable request.

Conflict of Interest

The authors declare that they have no financial, personal, or professional competing interests that could inappropriately influence, bias, or compromise the findings presented in this manuscript.

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